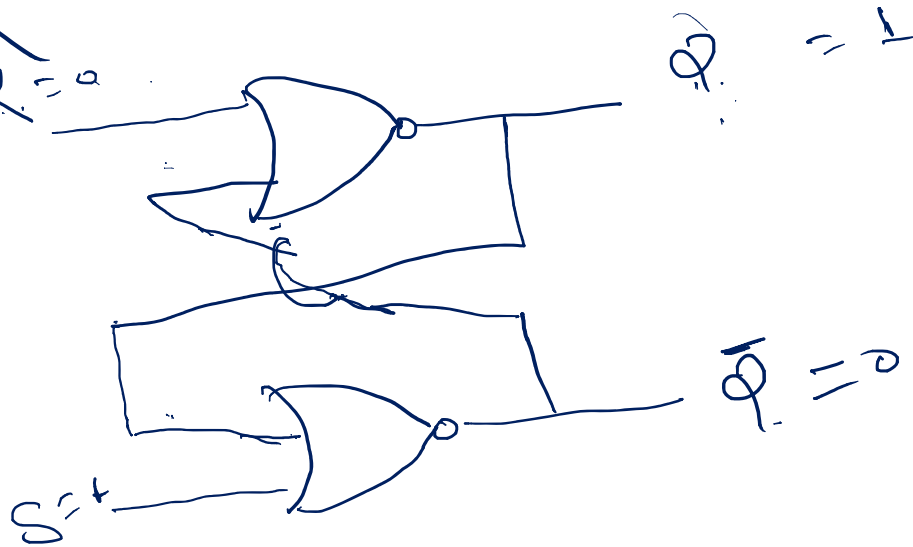


Sequential Circuits

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Latch! - SR Latch

NOR
 $R=0$



Reset $Q = 0$
~~Reset~~ $Q = 1$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	0

case $\Rightarrow$

$S=0$ $R=1$ $Q=0$ & $\bar{Q}=1$
 $S=0$ $R=0$ $Q=0$ $\bar{Q}=1$ memory

Case - $\Rightarrow$

$S=1$ $R=0$ $Q=1$ & $\bar{Q}=0$
 $S=0$ $R=0$ $Q=1$ & $\bar{Q}=0$ memory

case - II

Invalid / unused
state

$S = 1$

$S = 0$

$R = 1$

$R = 0$

X

$Q = 0$ & $\bar{Q} = 0$

$Q = 0$ & $\bar{Q} = 1$

$Q = 1$ & $\bar{Q} = 0$

S

0

0

1

1

R

0

1

0

1

Q

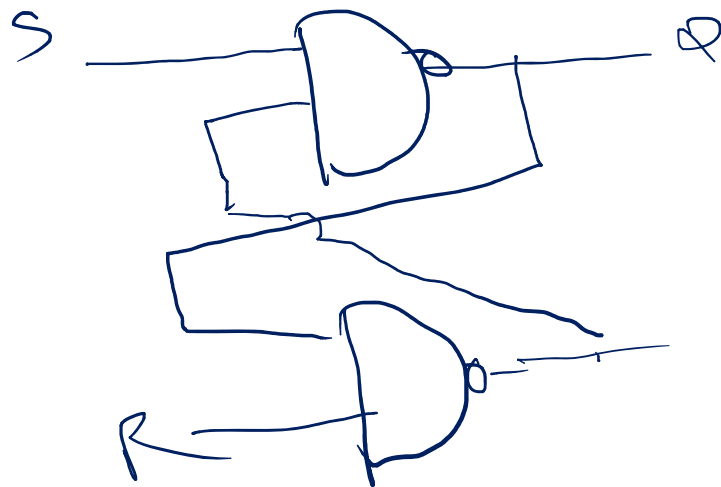
Q'

previous state

0

1

Invalid / unused

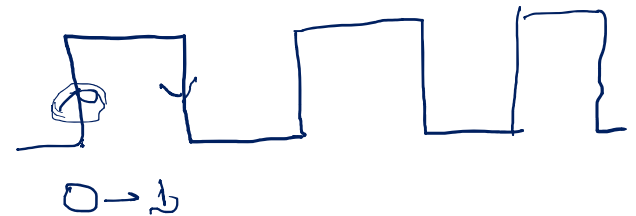
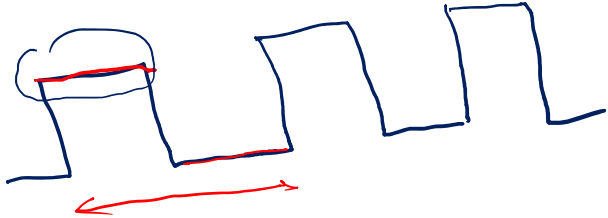
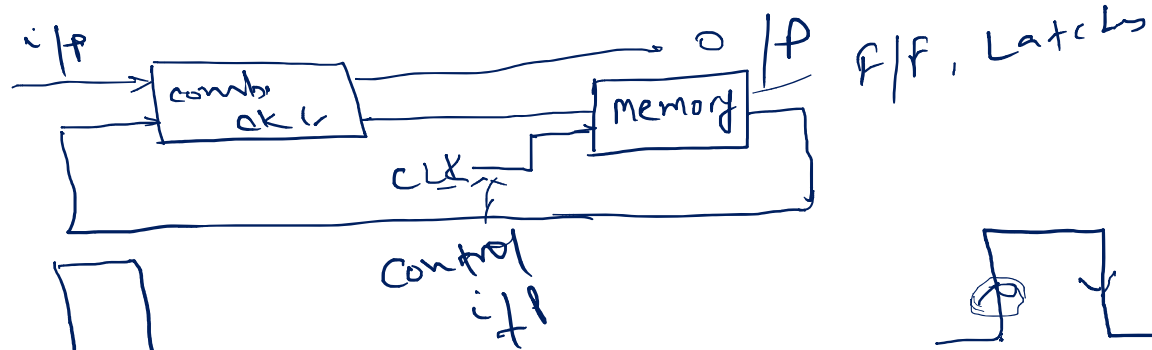


Q_1

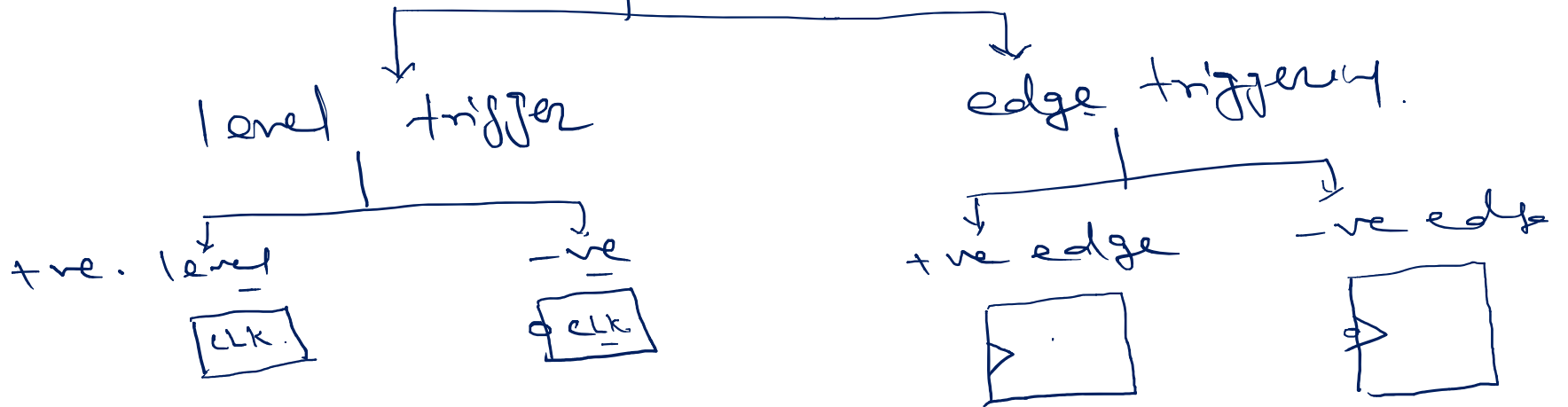
S	R
0	0
0	1
1	0
1	1

Q_1
Invalid / unused
1
0
Previous
state

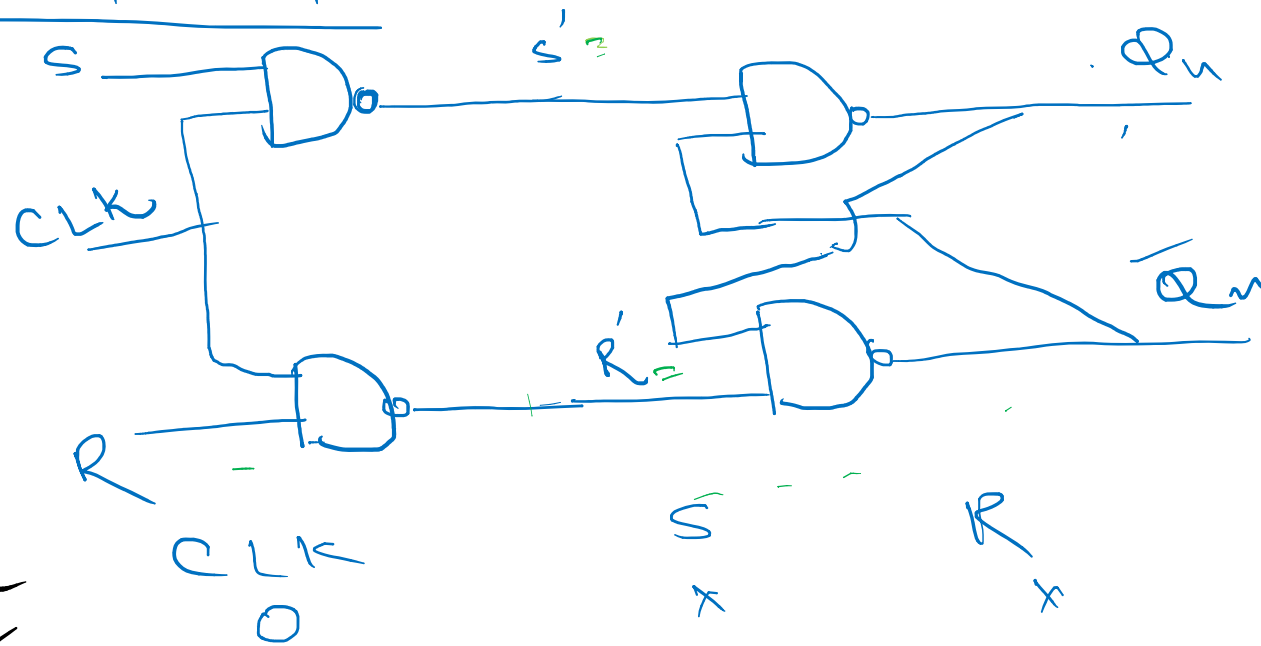
Triggering



Trigger



SR-Flip Flop



function
table

CLK
0
1
1
1
1

S	R
x	x
0	0
0	1
1	0
1	1

Q_{n+1}
Q_n
Q_n (hold)
0 (Reset)
1 (Set)
Invalid or unused state

characteristics table

S	R	Q_n	Q_{n+1}	
0	0	0	0	hold
0	0	1	1	
0	1	0	0	Reset
0	1	1	0	
1	0	0	1	set
1	0	1	1	
1	1	0	x	unused
1	1	1	x	

$$Q_{n+1} = \sum m(1, 4, 5) + \sum d(6, 7)$$

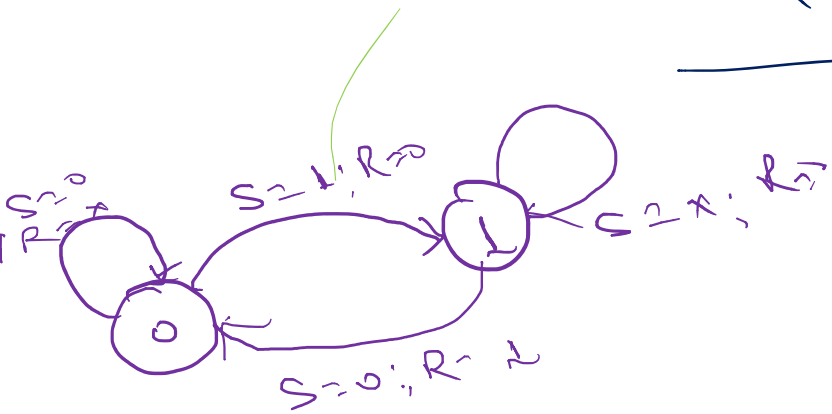
	RQ_n	$\bar{R}Q_n$	$\bar{R}\bar{Q}_n$	$R\bar{Q}_n$	RQ_n
Q_n	00	01	11	10	
S	1	1	1	1	1
R	1	1	1	1	1

$$Q_{n+1} = S + \bar{R}Q_n$$

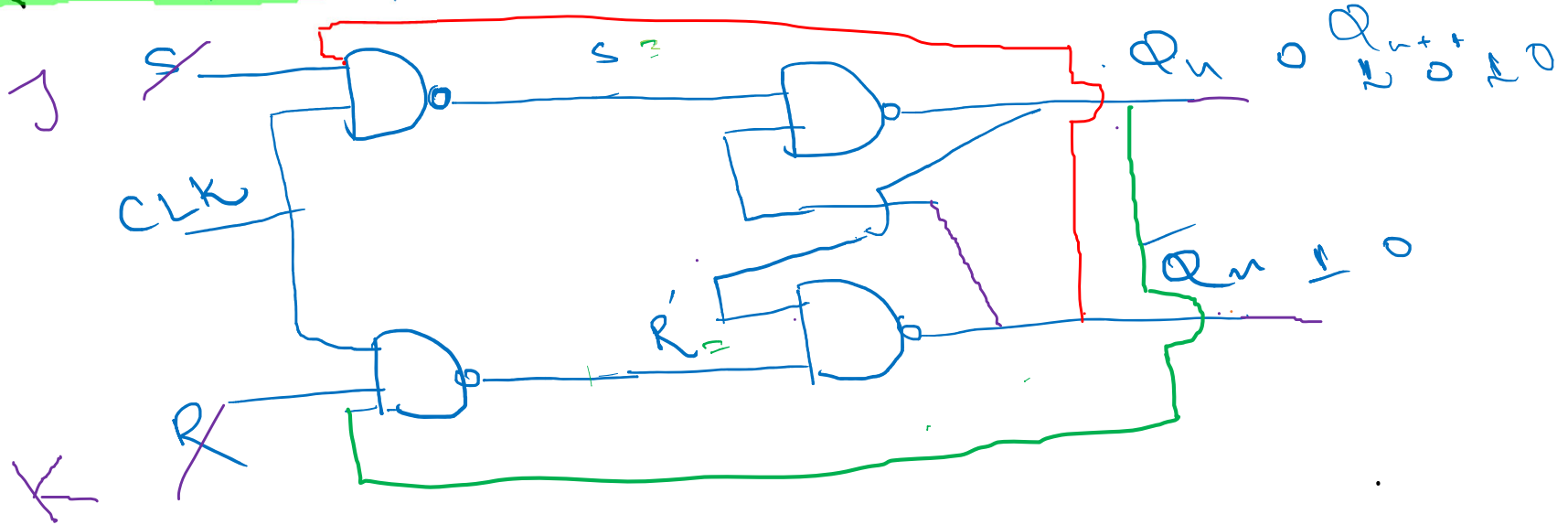
$$S, R \neq 1$$

Excitation table

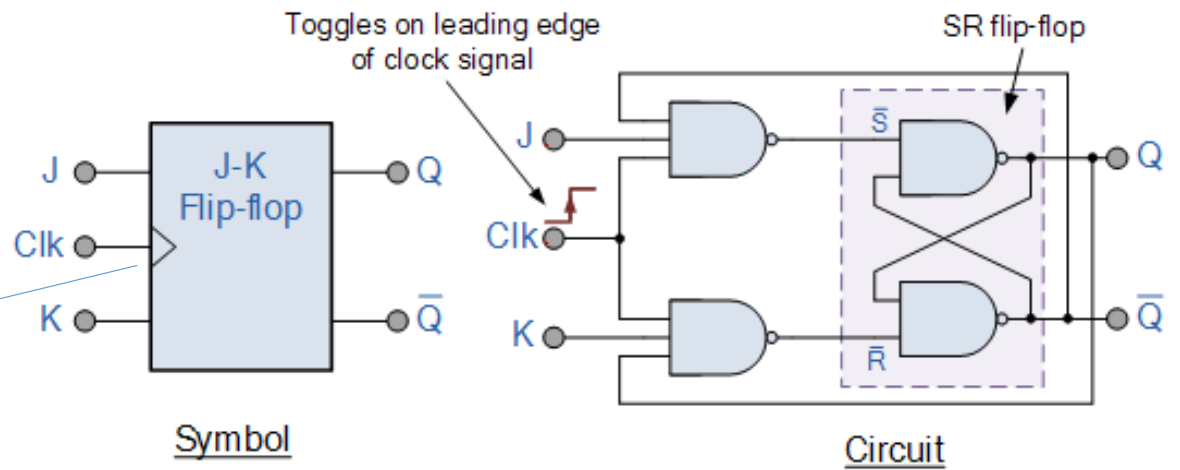
Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0



J-K flip flop

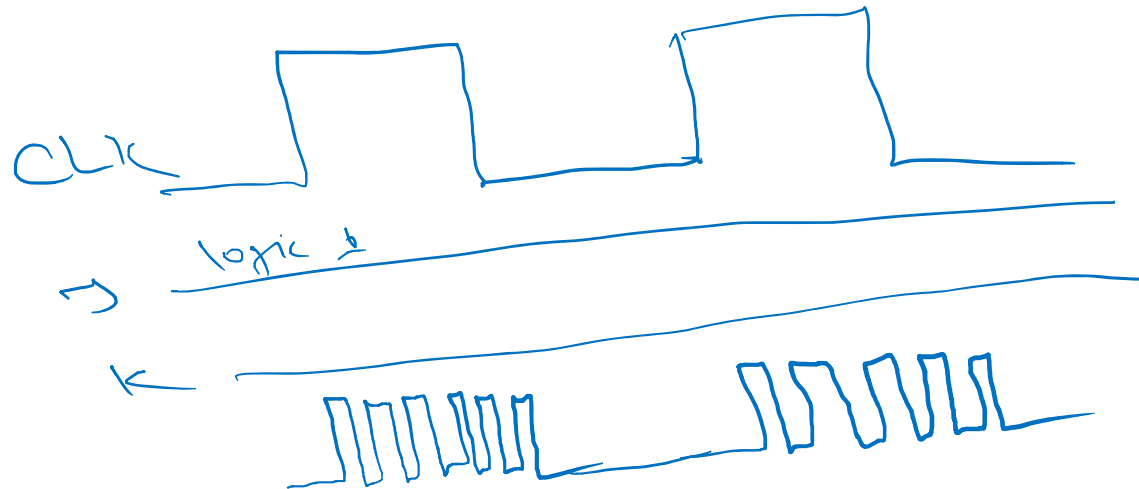


Positive Edge
Triggered FF



JK - flip flop

CLK	J	K	Q_{n+1}
0	X	X	Q_n
1	0	0	Q_n (hold)
1	0	1	0 (Reset)
1	1	0	1 (set)
1	1	1	$\bar{Q}_n$



Charac. table

$\bar{Q}_n, Q_n, \bar{Q}_n, Q_n$

0	0	1	0
1	0	1	1
0	1	0	0
1	1	0	1

J	K	Q_n
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Q_{n+1}
0
1
0
0
1
1
1
0

hold
reset
set
toggle

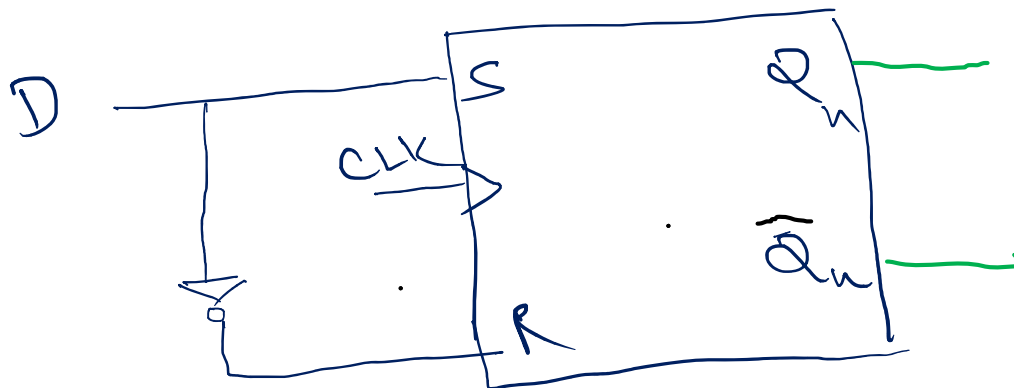
$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

$$Q_{n+1} = \sum m(1, 4, 5, 6)$$

Excitation table:-

Q_n	Q_{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

D - F / F



CLK

0

1

1

D

X

0

1

Q_{n+1}

Q_n.

0

1

- Reset

Charac. table

D	Q_n	Q_{n+1}	
0	0	0	
0	1	0	
1	0	1	
1	1	1	$Q_{n+1} = D$

Excitation table

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	1
1	1	0

T - f/p

truth table

CLK

0

1

1

T

X

0

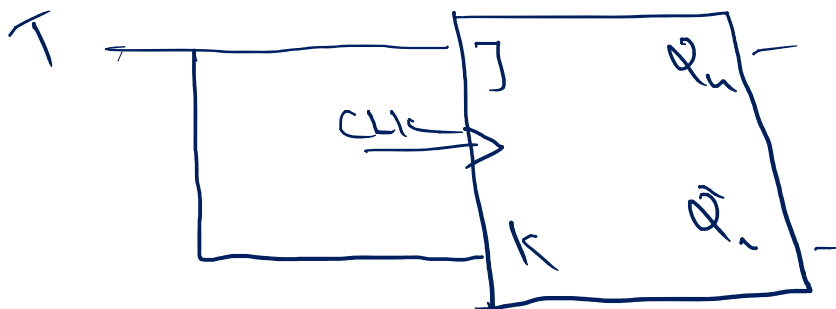
1

$Q_n + 1$

Q_n - memory

Q_n - hold

$\bar{Q}_n \rightarrow$ toggle



Charc. table

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

$$Q_{n+1} = T \oplus Q_n$$

Excitation table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Q_n	Q_{n+1}	S	R	J	K	D	T
0	0	0	x	0	x	0	0
0	1	1	0	1	x	1	1
1	0	0	1	x	1	0	1
1	1	x	0	x	0	1	0

$$Q_{n+1} = S + \bar{R}Q_n$$

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

$$Q_{n+1} = D$$

$$Q_{n+1} = T \oplus Q_n$$

Moore and Mealy State Machine

State Machine



Moore

1) O/P is function $\rightarrow$ Present State

2) Design is easy

3) Take more State

Mealy

O/P $\leftarrow$ Present State
Present I/P

Complex

less of state

Conversion of One Flip-flop to Other

1. Construct required Flip flop characteristics table
2. Fill the excitation table of available Flip Flop
3. Write the excitation equation
4. Minimize the expression
5. Implement the circuit

JK to D F/F

D
 $1-00$
 Q_n
 0
 1
 0
 A
 Q_{n+1}
 0
 0
 1
 0

D Q_n Q_n Q_n

$\bar{D}$		0	x
D		1	x

$J = D$

K-MAB
for $\rightarrow$

Q_n
 0
 x
 1
 x
 x
 Q_{n+1}
 x
 1
 x
 0

K-MAB
for $\rightarrow$

D Q_n Q_n Q_n

$\bar{D}$		x	1
D		x	

$K = \bar{D}$

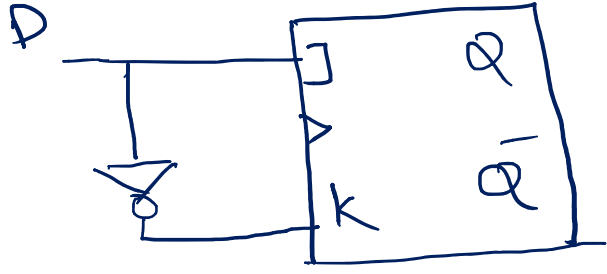
JK to xz

J	K	0	0	X
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

Q_{n+1}

Q_n	0	1	0	1	0	1
Q_{n+1}	0	0	1	0	1	1

Arrows indicate transitions from Q_n to Q_{n+1} .



1 1 0 1 0 1 0

1 0 1 0 1 0 1

	$yQ_n \bar{y}Q_n$	$\bar{y}Q_n$	yQ_n	$\bar{y}Q_n$
x	x_0	x_1	x_3	x_2
x	x_4	x_5	x_7	x_6

$$y = \bar{x}$$

	yQ_n	$\bar{y}Q_n$	$\bar{y}Q_n$	$\bar{y}Q_n$
x	x_0	x_1	x_3	x_2
x	x_4	x_5	x_7	x_6

$$x = y$$

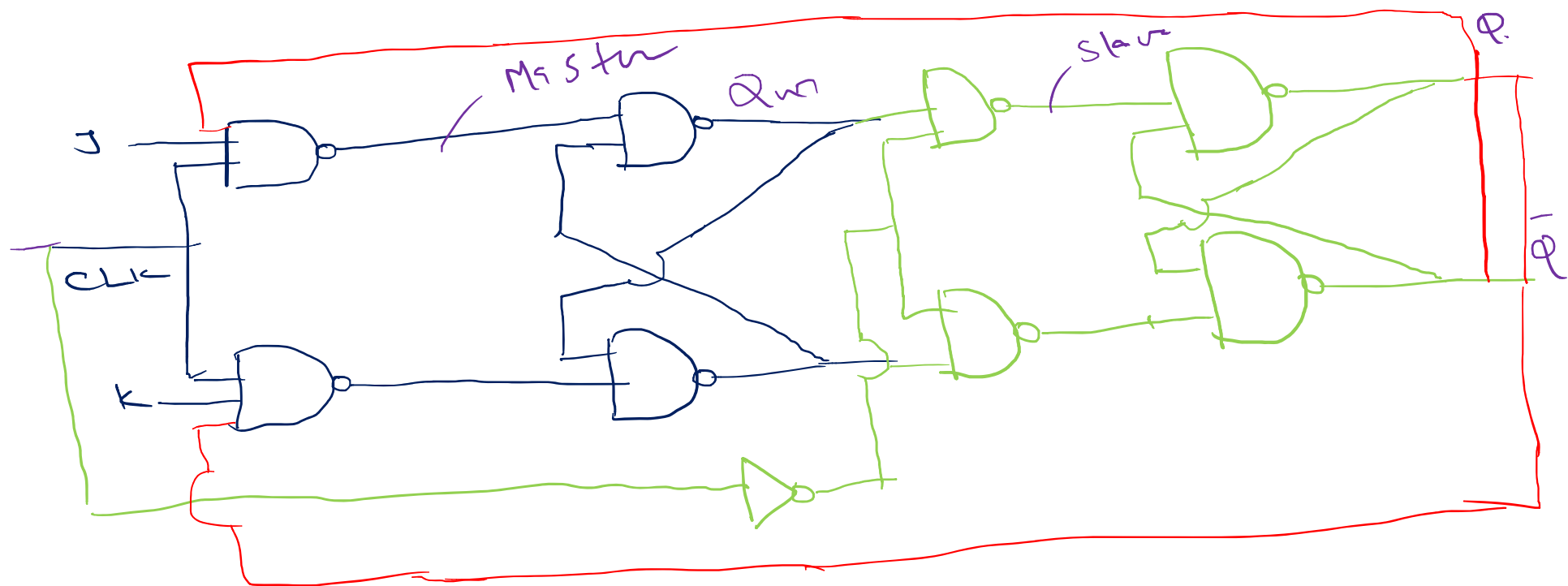
* Race-around conditions *

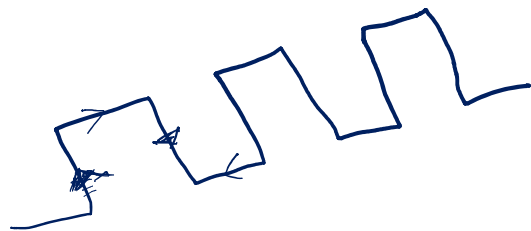
Conditions to avoid Race-around

1) $t_{pw} < t_{pd} \text{ ff} < T_{clk}$

2) edge trigger

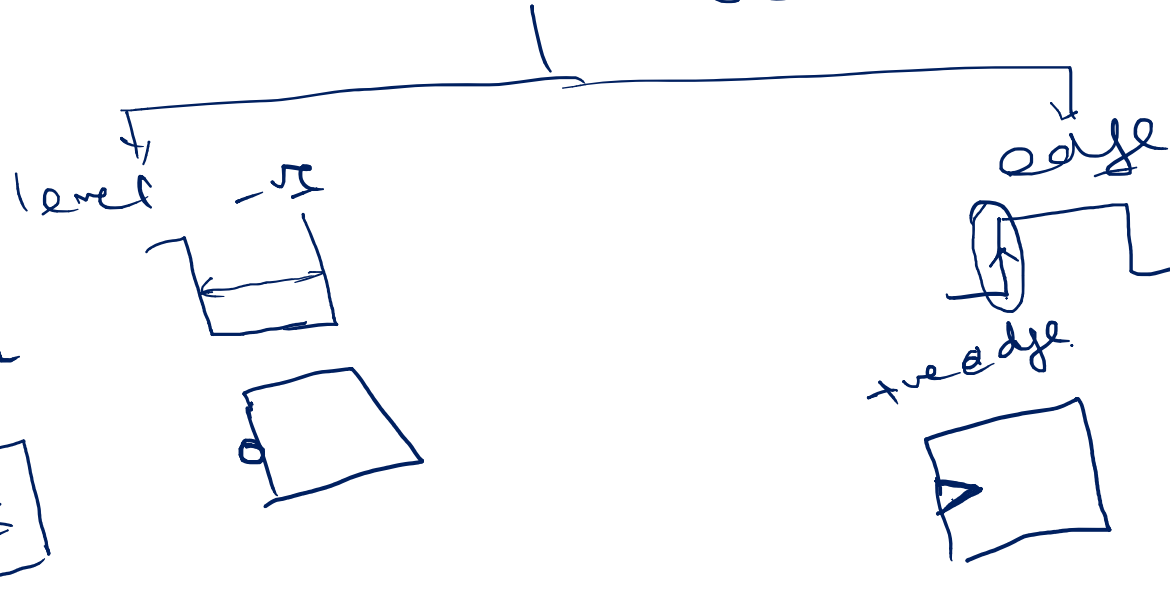
3) Master-slave ff is used





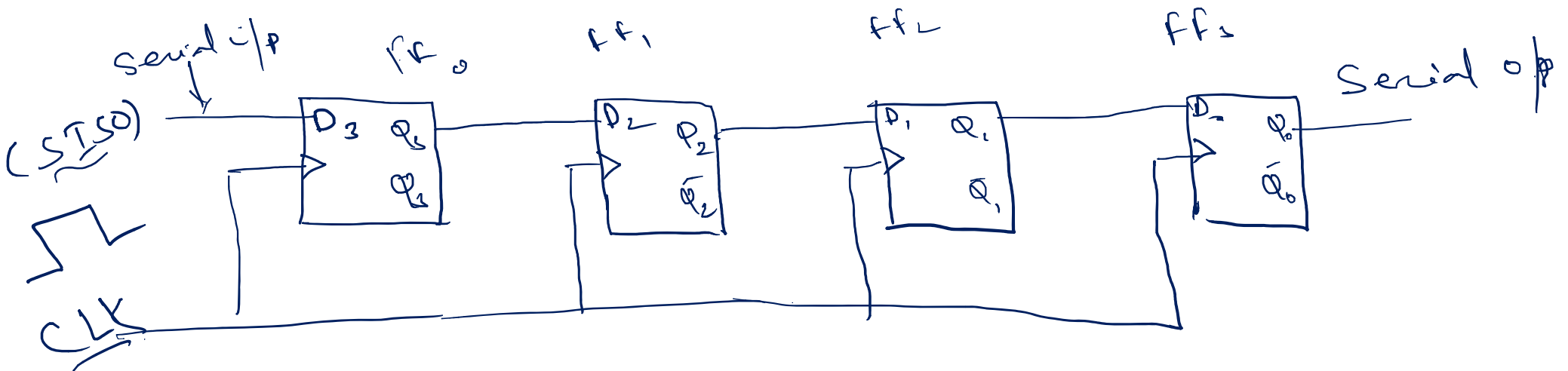
Shift Register

clock trigger.



true





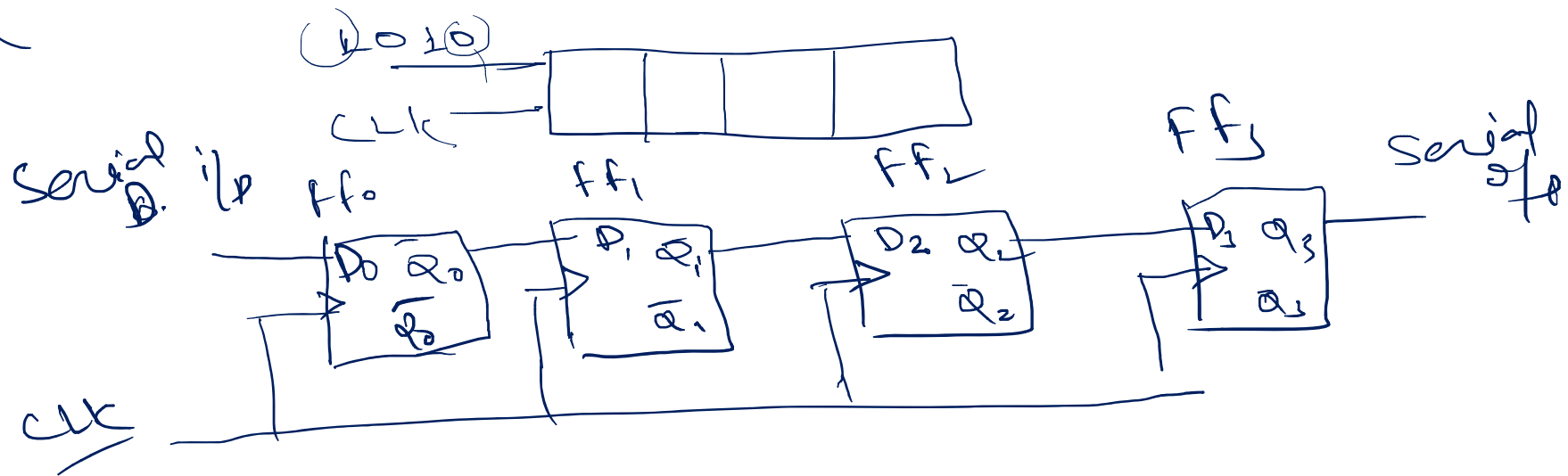
* Shift register is unidirectional (left to right)

SISO (Serial In Serial Out)



CLK

CLK	Data	ϕ_3	ϕ_2	ϕ_1	ϕ_0
0	1 0 1 1	0	0	0	0
1		1	0	0	0
2		1	1	0	0
3		0	1	1	0
4		1	0	1	1



Shift Register

SISO :- In SISO register to provide n bit data Serial In, CLK require minimum n clk pulses.

→ For n bit regⁿ to the i/p data. it will provide $n \times T_{clk}$ delay

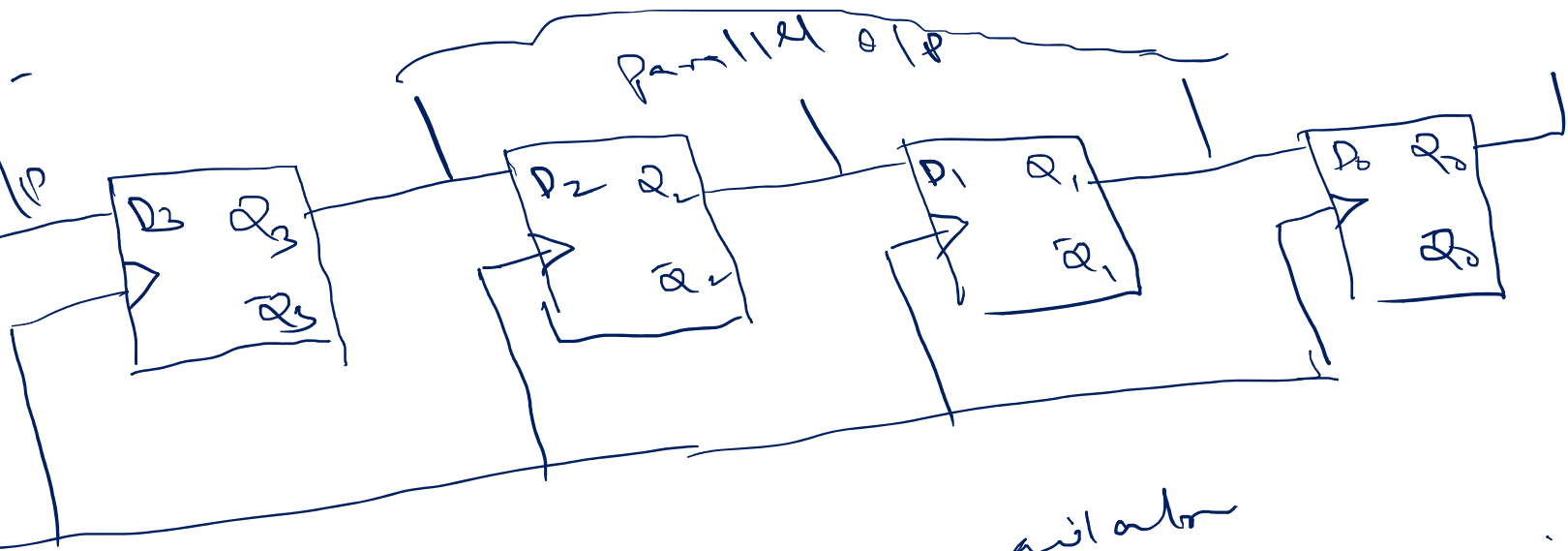
→ In n -bit SISO register to provide n bit Serial output it require minimum $n-1$ CLK pulses

SISO :-

Serial i/p

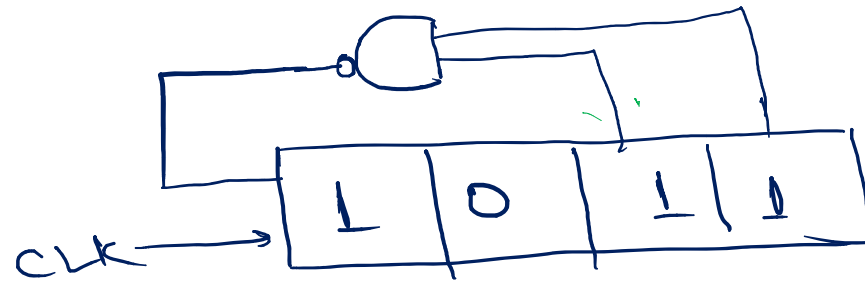
clk

Parallel o/p



- o/p of each stages are available
- In SISO to provide serial i/p it requires
 ~ clk pulse and to provide parallel o/p
 ~ clk pulse are reqd.
- It converts Serial data to Parallel
 Temporal code $\Rightarrow$ Spatial Code

Q. 1.



S190

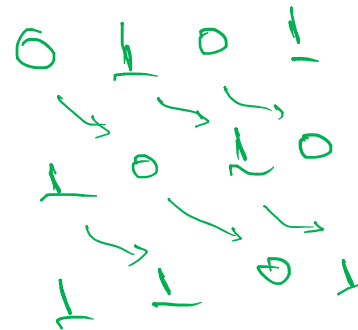
After 3rd CLK Pulse, the data ^{Fig 1} presenting in Shift register is -----

Solⁿ :-

1st CLK pulse

2nd

3rd

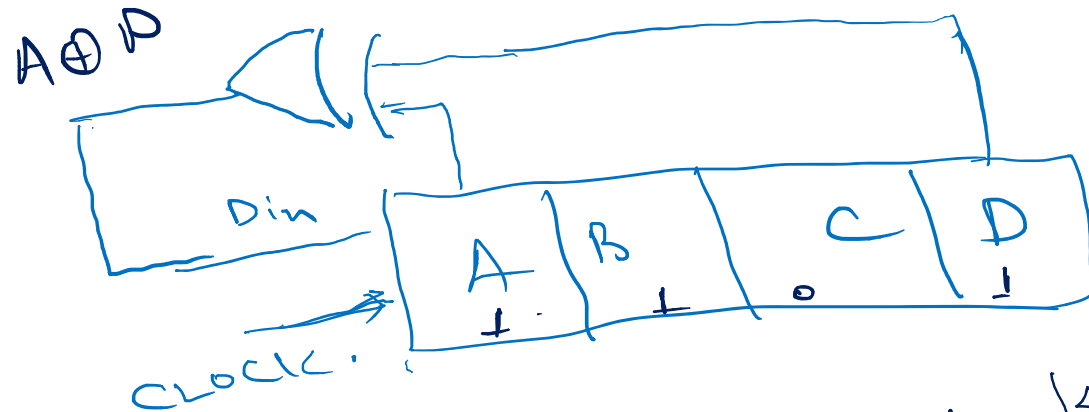


Q.2 $D_{in} \rightarrow A$; $A \rightarrow B$ $B \rightarrow C$ $C \rightarrow D$

$ABCD = 1101$

no. of CLK pulses are reqd to reach the State.

$ABCD = 1111$



Initial

After 1^{st}

2
3

1	1	0	1
0	1	1	0
0	0	1	1
1	0	0	1

4	0	0	0
5	0	0	0
6	0	0	0
7	1	0	0
8	1	1	0
9	1	1	0
10	1	1	0
11	1	1	0
12	1	1	0
13	1	1	0
14	1	1	0
15	1	1	0

PISO (Parallel In serial out)

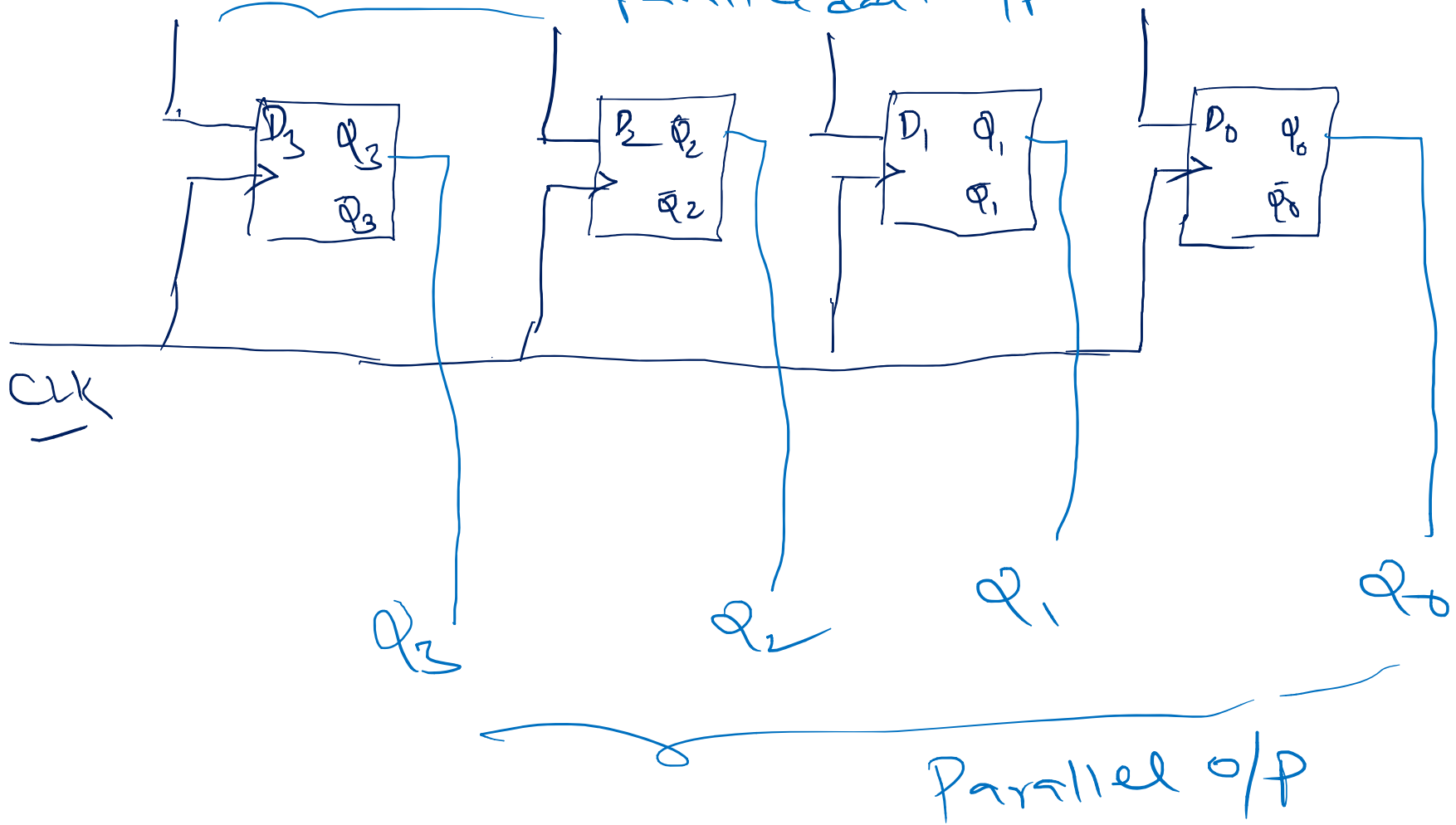


PI - one CLK
S.o/p - (n-1) CLK

Control = 0 (PI)
Control = 1 (Serial o/p)

$$Y = S_0 I_0 + S_1 I_1$$

PIPO (Parallel In Parallel out)
parallel data i/p ~



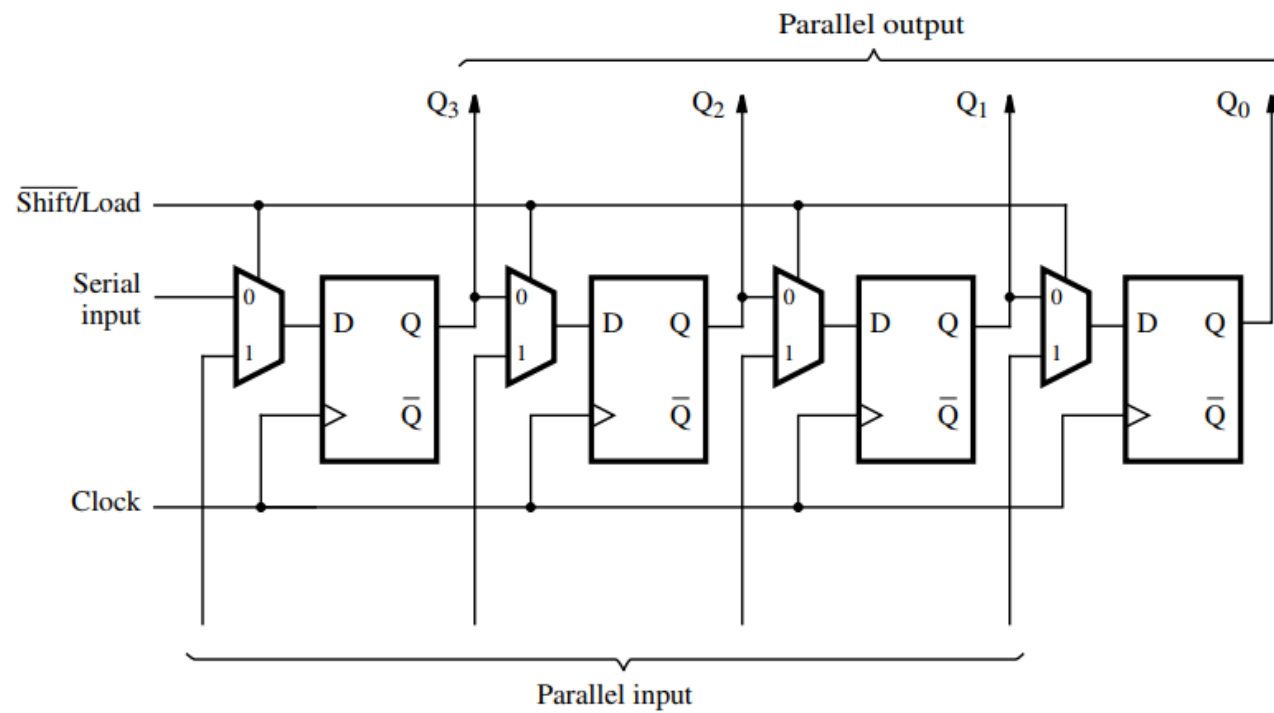


Figure 1 Parallel-access shift register

Figure 1 shows a four-bit shift register that provides the parallel access. A 2-to-1 multiplexer on its D input allows each flip-flop to be connected to two different sources. One source is the preceding flip-flop, which is needed for the shift-register operation. The other source is the external input that corresponds to the bit that is to be loaded into the flip-flop as a part of the parallel-load operation. The control signal *Shift/Load* is used to select the mode of operation. If *Shift/Load* = 0, then the circuit operates as a shift register. If *Shift/Load* = 1, then the parallel input data are loaded into the register. In both cases the action takes place on the positive edge of the clock. In Figure 1 we have chosen to label the flip-flops' outputs as $Q_3, \dots, Q_0$ because shift registers are often used to hold binary numbers. The contents of the register can be accessed in parallel by observing the outputs of all flip-flops. The flip-flops can also be accessed serially, by observing the values of Q_0 during consecutive clock cycles while the contents are being shifted. A circuit in which data can be loaded in series and then accessed in parallel is called a series-to-parallel converter. Similarly, the opposite type of circuit is a parallel-to-series converter. The circuit in Figure 1 can perform both of these functions.

	$u/p(u)$	o/p
SISO	η	$u-b$
SIDO	η	0
PI SO	$\downarrow$	$u-L$
PI DO	$\downarrow$	0

* COUNTERS *

- * Counters are basically used to count
- 1) To count no of clk pulses applied.
 - 2) Freq. divider
 - 3) Waveform generat.
 - 4) Timers
 - 5) Frequency Measurement
 - 6) Range/Distance Measurement
 - 7) Pulse width Measurement.

1 ff - 2 states (2^1)
 2 ff - 4 states (2^2)
 3 ff - 8 states (2^3)
 "
 n ff - 2^n states

* In counter with n-f/f max^m possible states are 2^n

n → No. of flip flops

$$N \leq 2^n$$

N = No. of states

$$n \geq \log_2 N$$

* No. of states
 used in counter is
 known as Modulus
counter (Mod n₀)
 Mod s counter.
s states

Counters



Asynchronous.

- 1) Different F/F are applied with different CLK pulse
- 2) slower
- 3) only fixed seq $\begin{matrix} \swarrow \text{up} \\ \searrow \text{down} \end{matrix}$
- 4) Decoding errors
Ripple counter

Synchronous counters

All F/F are applied with same CLK pulse

Faster

- 3) It provides the count sequence.

4) No Decoding error

Ring counter Johnson counter

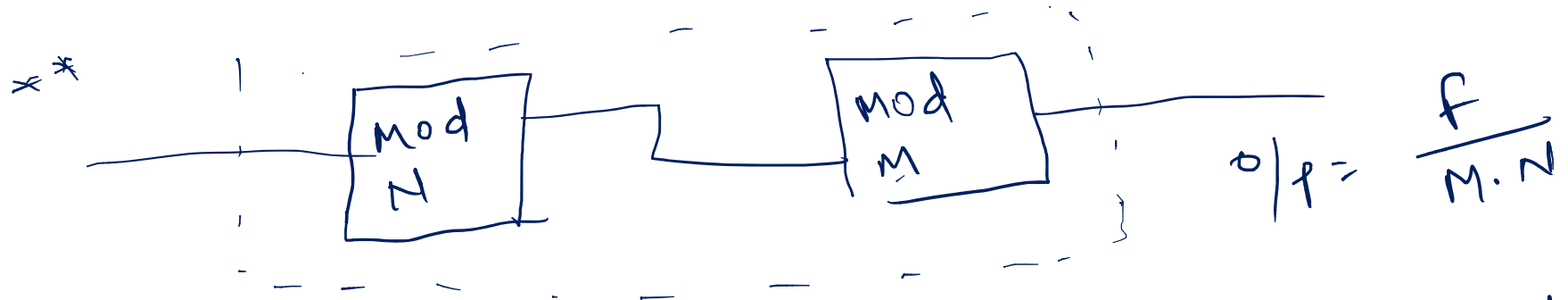
No. of states $\leftarrow N \leq 2^n \leftarrow \text{no. of } f \text{ | } f$

→ No. of states used in the counter is known as Modulus counter (Mod No)
 Mod 5 Counter → 5 states



* A decade counter is applied with CLK having freq 10 MHz. The o/p freq will be ---

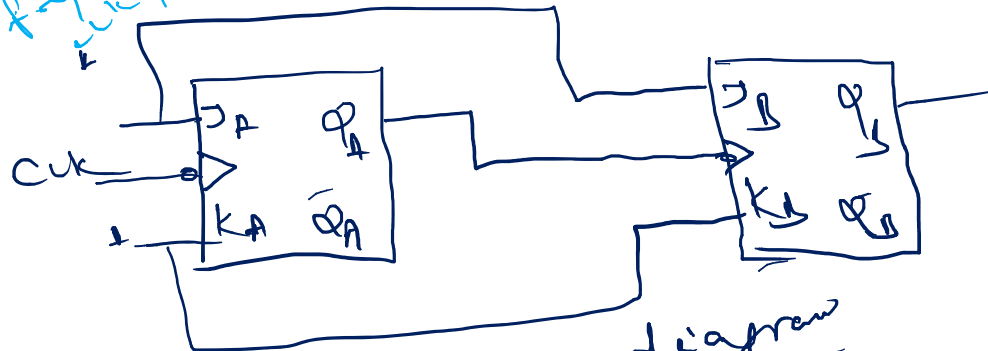
$$f = \frac{10 \text{ MHz}}{10} = 1 \text{ MHz}$$



* If Mod M counter is cascade with Mod N counter, then it will acts as a Mod MN counter

Ripple Counter

$f \Rightarrow$ freq. of input

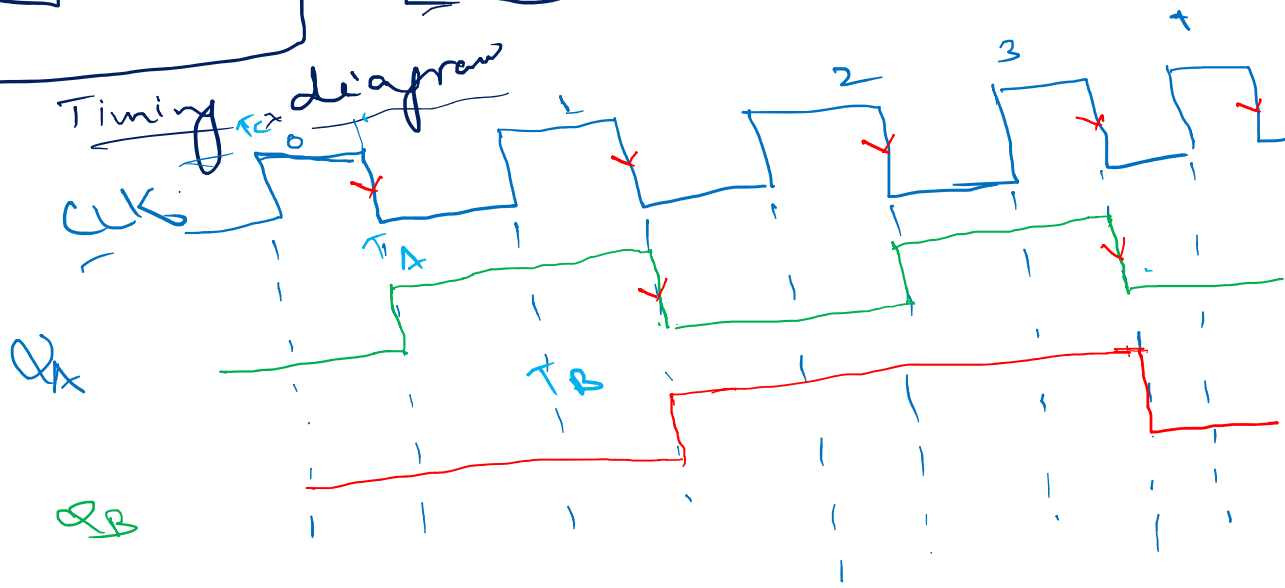


$f/4$

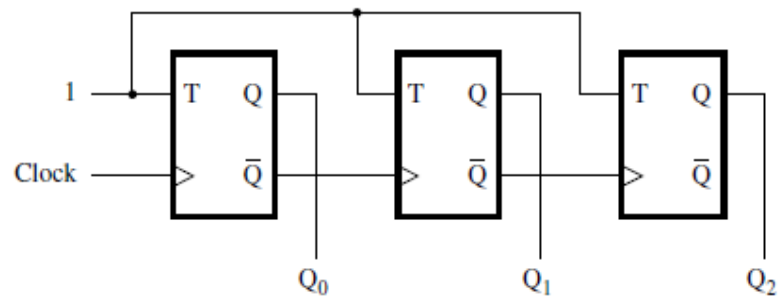
Timing diagram

Ck
0
1
2
3

Q_A
0
0
1
1

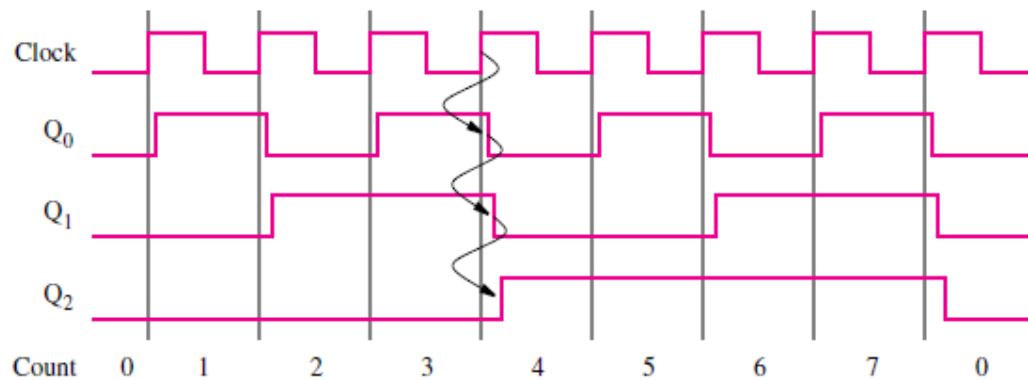


Up-Counter with T Flip-Flops



(a) Circuit

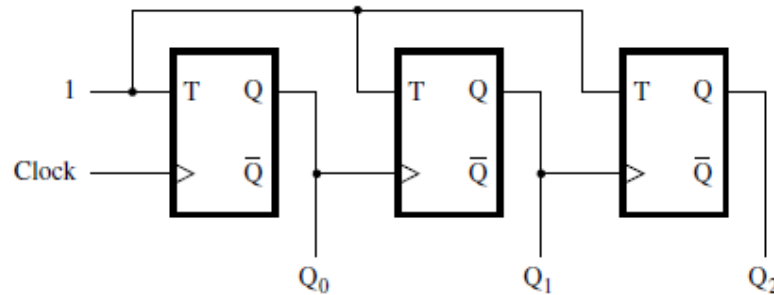
The clock inputs of the three flip-flops are connected in cascade. The T input of each flip-flop is connected to a constant 1, which means that the state of the flip-flop will be reversed (toggled) at each positive edge of its clock. The clock input of the first flip-flop is connected to the Clock line. The other two flip-flops have their clock inputs driven by the Q' output of the preceding flip-flop. Therefore, they toggle their state whenever the preceding flip-flop changes its state from $Q = 1$ to $Q = 0$, which results in a positive edge of the Q' signal.



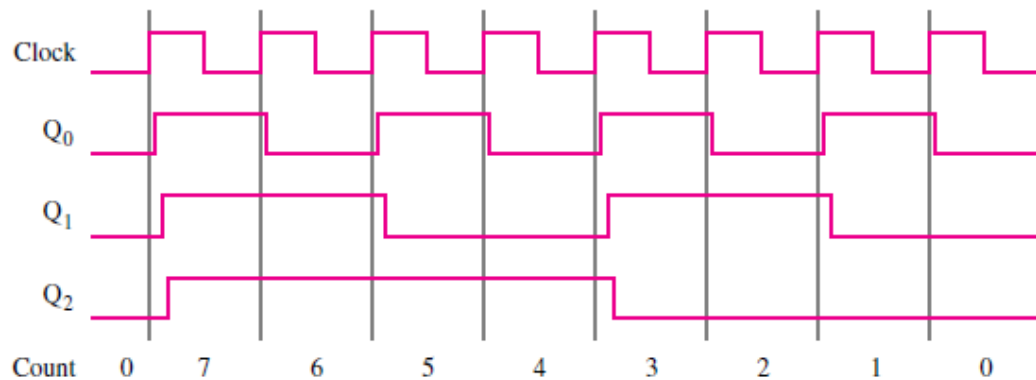
(b) Timing diagram

- ❖ The value of Q_0 toggles once each clock cycle. The change takes place shortly after the positive edge of the *Clock* signal.
- ❖ The delay is caused by the propagation delay through the flip-flop. Since the second flip-flop is clocked by Q_0' , the value of Q_1 changes shortly after the negative edge of the Q_0 signal.
- ❖ Similarly, the value of Q_2 changes shortly after the negative edge of the Q_1 signal.
- ❖ If we look at the values $Q_2Q_1Q_0$ as the count, then the timing diagram indicates that the counting sequence is 0, 1, 2, 3, 4, 5, 6, 7, 0, 1, and so on.

Down-Counter with T Flip-Flops



(a) Circuit

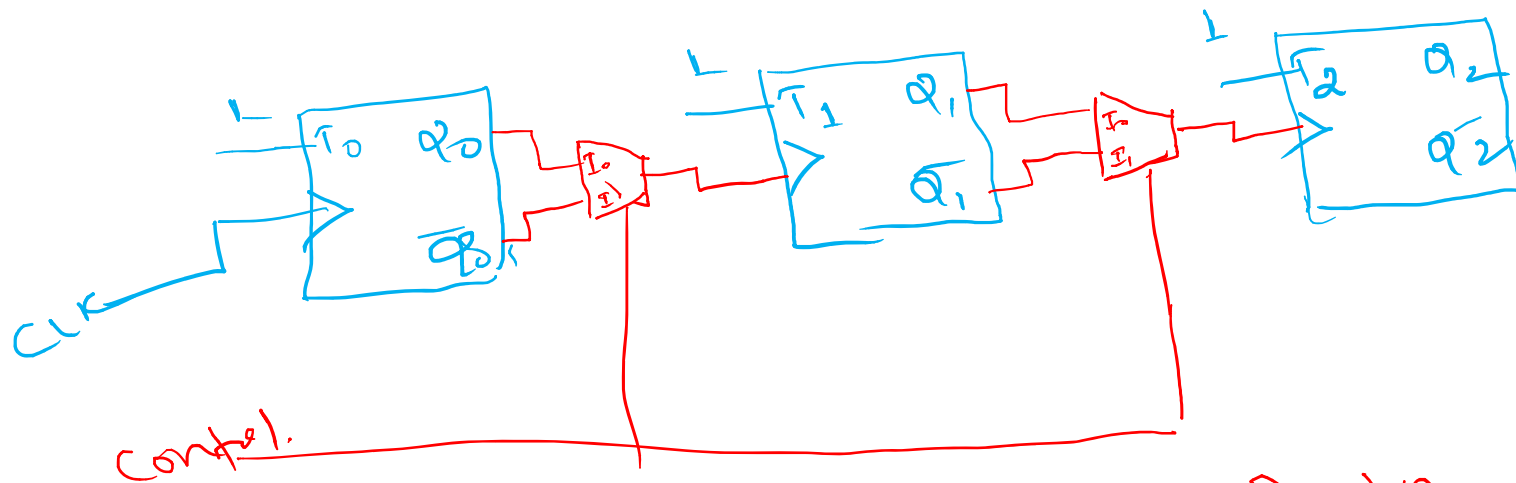


(b) Timing diagram

The only difference is that in Figure a the clock inputs of the second and third flip-flops are driven by the Q outputs of the preceding stages, rather than by the Q' outputs. The timing diagram, given in Figure b, shows that this circuit counts in the sequence 0, 7, 6, 5, 4, 3, 2, 1, 0, 7, and so on. Because it counts in the downward direction, we say that it is a *down-counter*.

A three-bit down-counter

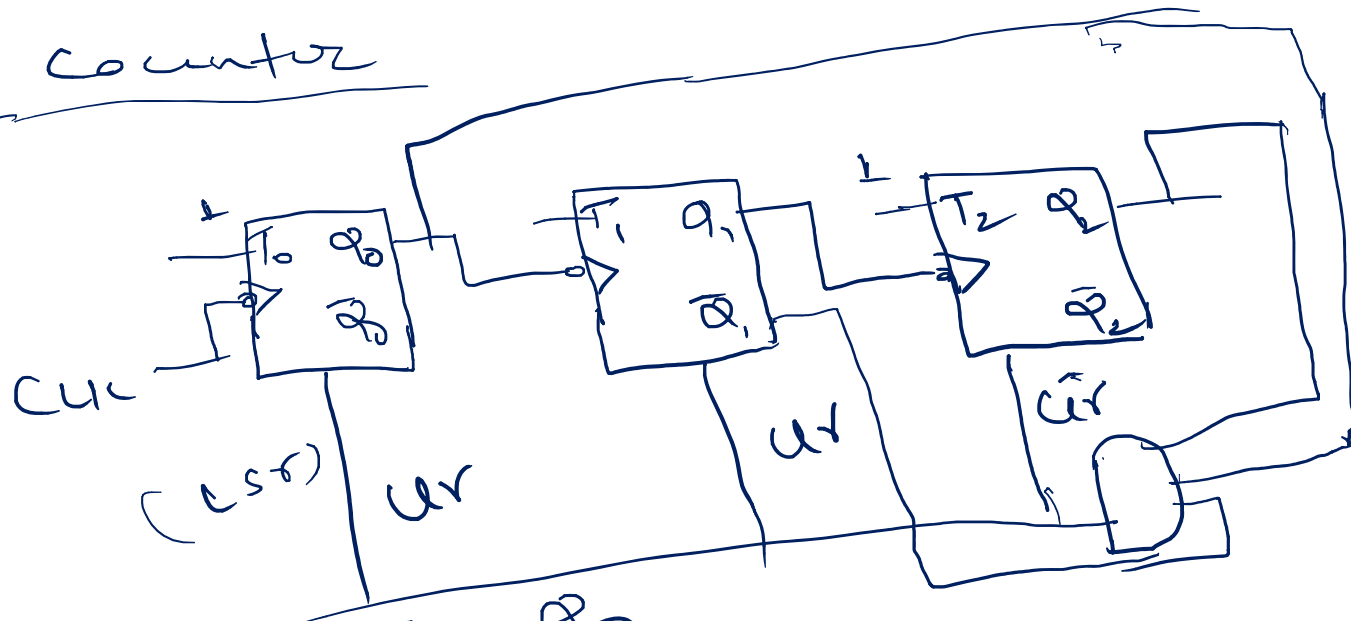
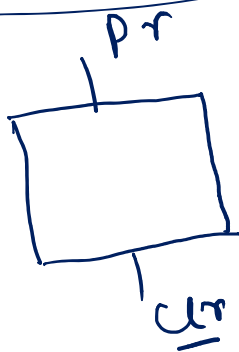
Ripple up/down Counter



control = 0 → DOWN Counter
 control = 1 → UP Counter

Mod 5 Ripple Counter

Q_2, Q_1, Q_0
 101
 Q_2, Q_1, Q_0

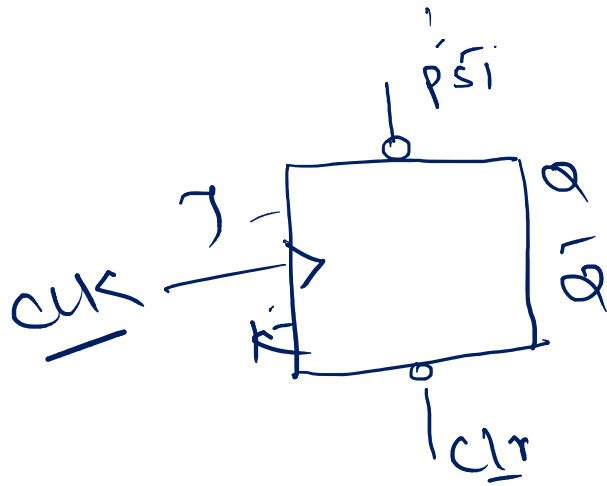


clk
 0
 1
 2
 3
 4
 5

Q_2
 0
 0
 0
 0
 1
 0

Q_1
 0
 0
 0
 0
 1
 0

Q_0
 0
 0
 1
 0
 1
 0



$P_{ST} = 0$
 $u_r = 0$
 P_{ST}
 0
 1
 1
 0

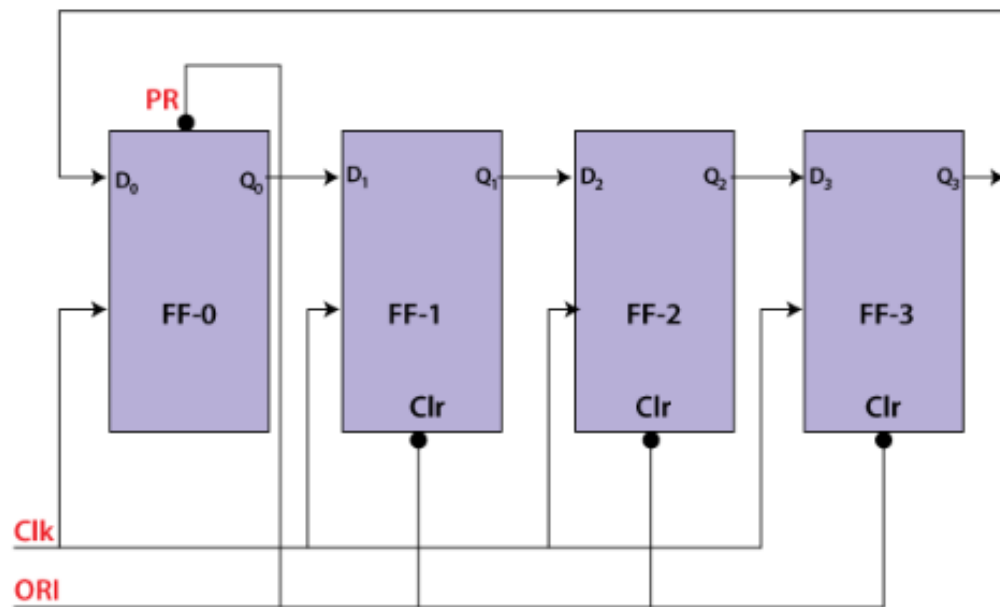
$\Rightarrow Q_1 = 1$
 $\Rightarrow Q_2 = 0$
 C_{lr} ϕ
 1 1
 0 0
 1 0
 0 0
 Normal.
 Not used

Ring Counter

A ring counter is a special type of application of the Serial IN Serial OUT Shift register. The only difference between the shift register and the ring counter is that the last flip flop outcome is taken as the output in the shift register. But in the ring counter, this outcome is passed to the first flip flop as an input. All of the remaining things in the ring counter are the same as the shift register. The **Overriding input(ORI)** is used to design this circuit. The Overriding input is used as **clear** and **pre-set**.

1.PR = 0, Q = 1

2.CLR = 0, Q = 0



No. of states in Ring counter = No. of flip-flop used

ORI	CLK	Q0	Q1	Q2	Q3
low	X	1	0	0	0
high	low	0	1	0	0
high	low	0	0	1	0
high	low	0	0	0	1
high	low	1	0	0	0

* No. of unused state = $2^n - n$

* In n -bit ring counter if i/p frequency is f . then the o/p freq will be f/n

→ In ring counter phase shift betⁿ the successive waveform is $\frac{360}{n}$

Application

1) Stepper motor - control

2) SAR type ADC

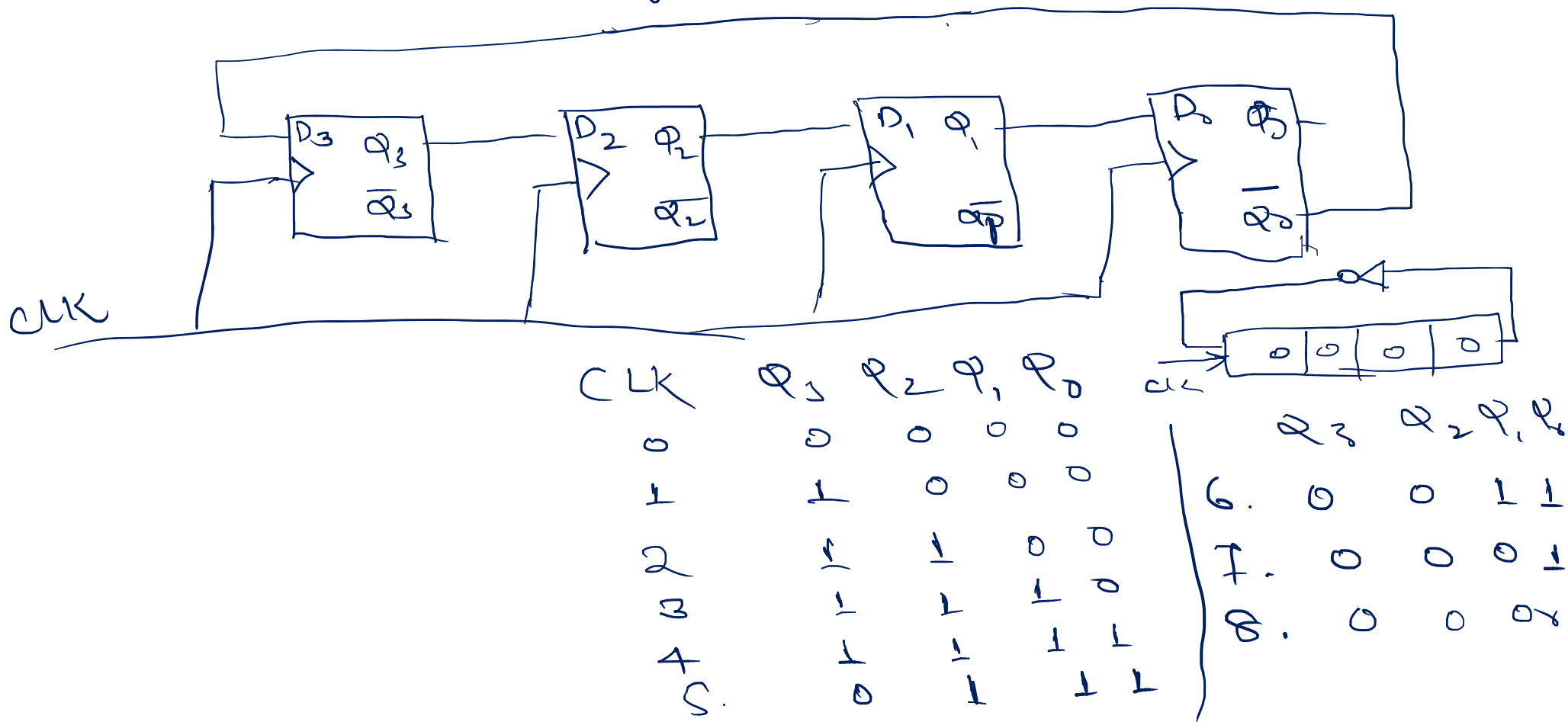
3) waveform generation

$$T_{clk} \geq t_{pd} F_f$$

$$f_{clk} \leq \frac{1}{t_{pd} - F_f}$$

$$f_{max} = \frac{1}{t_{pd} - F_f}$$

Johnson counter (switch tail counter)
or twisted Ring counter



4 FF $\rightarrow$ 8 states.

n FF $\rightarrow$ 2^n states

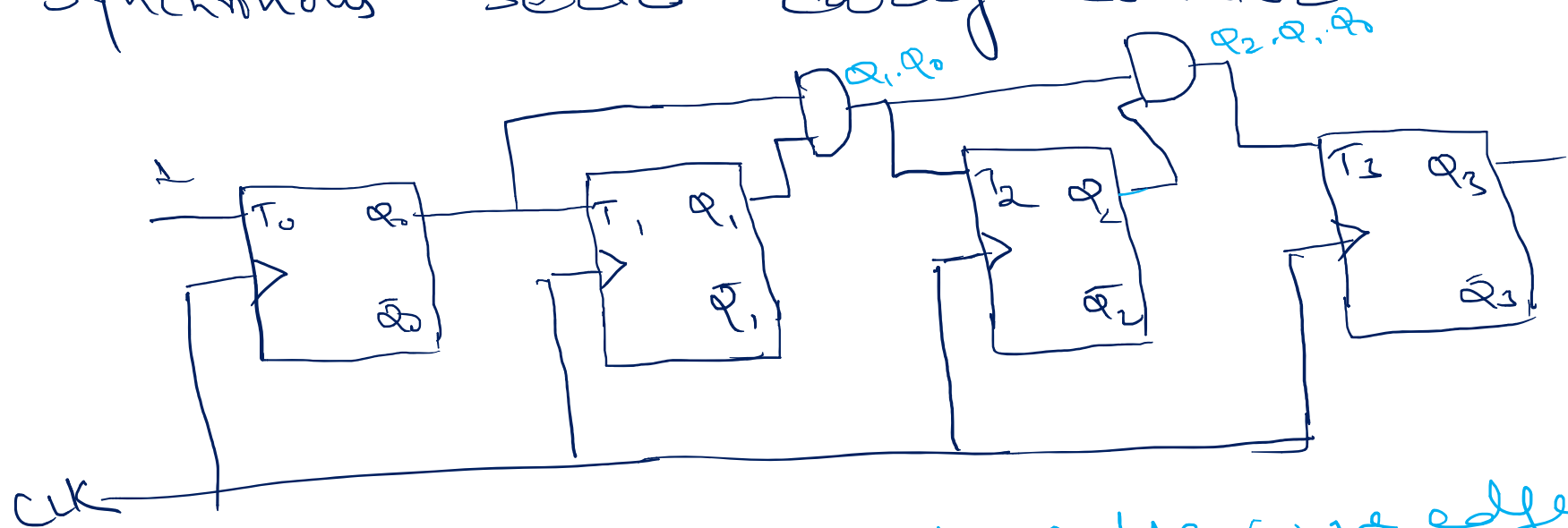
* No. of unused state = $2^n - 2^n$

*
$$\underline{O/p \text{ Freq}} = \underline{f / 2^n}$$

Another name.

- $\rightarrow$ switch tail counter $\sim$ Mobius counter
- $\rightarrow$ Creeping counter
- $\rightarrow$ walky counter

* Synchronous Series Carry Counter



- * Q_0 toggles for every clk pulse (+ve edge trigger)
 - * Q_1 " " " " " " " " " " " "
 - * Q_2 " " " " " " " " " " " "
 - * Q_3 " " " " " " " " " " " "
- when $Q_0 = 1$
- when $Q_1, Q_0 = 1$
- when $Q_2, Q_1, Q_0 = 1$

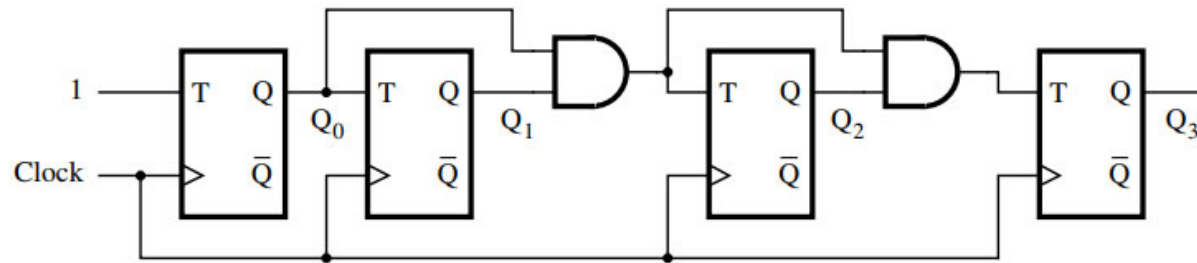
CLK	Q ₃	Q ₂	Q ₁	Q ₀
0	0	0	0	1
1	0	0	1	0
2	0	0	1	1
3	0	0	1	0
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
...	...	...	...	...
15	1	1	1	1

$$T_{CLK} \geq t_{pd-FF} + (n-2)t_{pd-AND}$$

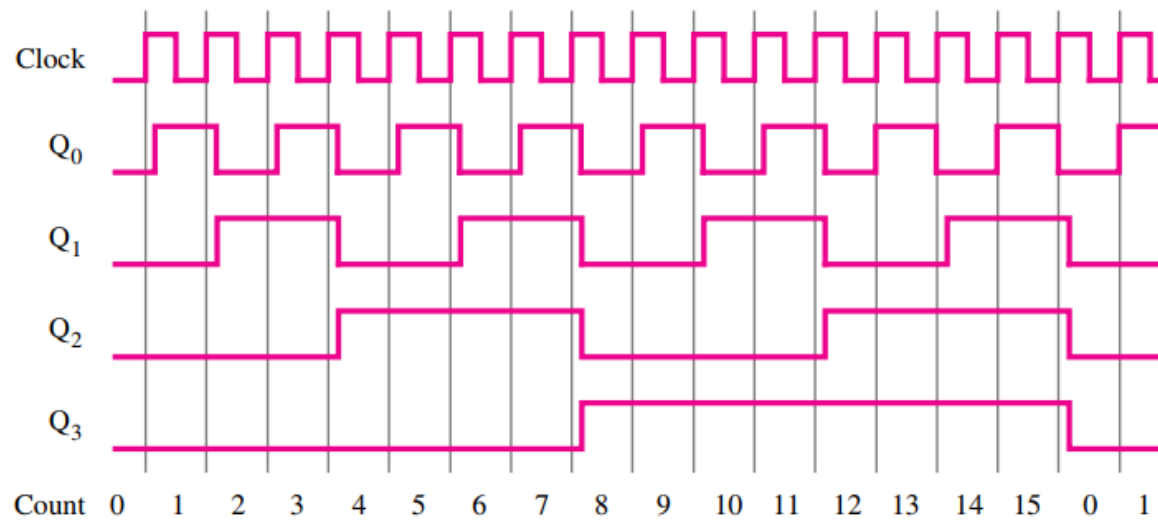
$$f_{CLK} \leq \frac{1}{t_{pd-FF} + (n-2)t_{pd-AND}}$$

$$f_{max} = \frac{1}{t_{pd-FF} + (n-2)t_{pd-AND}}$$

Synchronous Counter with T Flip-Flops



(a) Circuit



(b) Timing diagram

Figure A four-bit synchronous up-counter

Table 5.1 shows the contents of a three-bit up-counter for eight consecutive clock cycles, assuming that the count is initially 0. Observing the pattern of bits in each row of the table, it is apparent that bit Q_0 changes on each clock cycle. Bit Q_1 changes only when $Q_0 = 1$. Bit Q_2 changes only when both Q_1 and Q_0 are equal to 1. In general, for an n -bit up-counter, a given flip-flop changes its state only when all the preceding flip-flops are in the state $Q = 1$. Therefore, if we use T flip-flops to realize the counter, then the T inputs are defined as

Table 5.1 Derivation of the synchronous up-counter.

Clock cycle	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

$$T_0 = 1$$

$$T_1 = Q_0$$

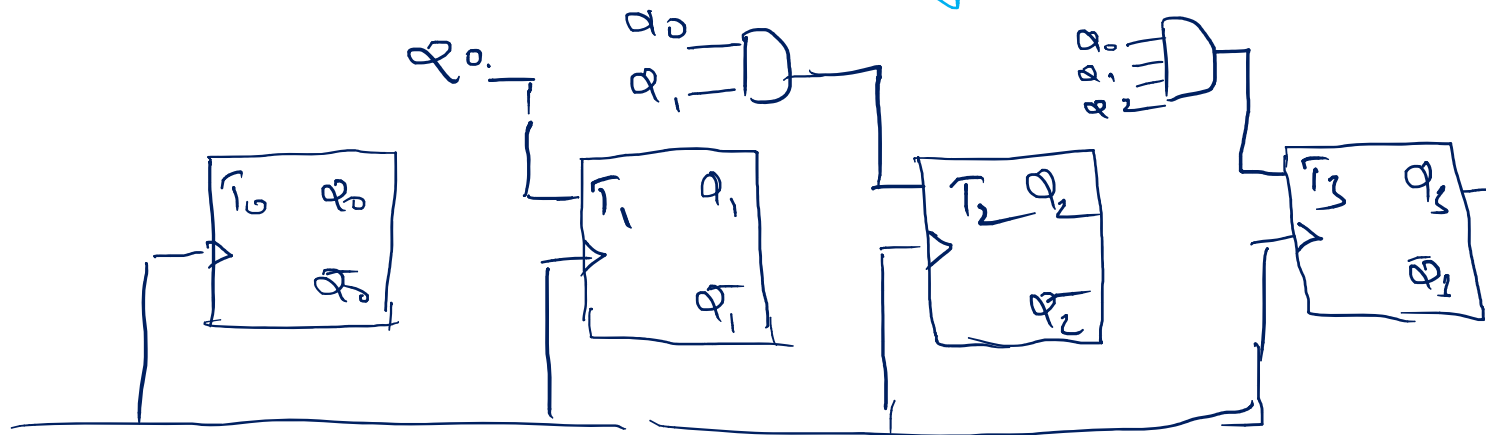
$$T_2 = Q_0 Q_1$$

$$T_3 = Q_0 Q_1 Q_2$$

...

$$T_n = Q_0 Q_1 \cdots Q_{n-1}$$

Synchronous Parallel Carry Counter:-



$$T_{CLK} \geq t_{pd-FF} + t_{pd-AND}$$

$$f_{CLK} \leq \frac{1}{t_{pd-FF} + t_{pd-AND}}$$

$$f_{max} = \frac{1}{t_{pd-FF} + t_{pd-AND}}$$



Design of Mod – N synchronous Counter

Design for Mod-N counter :

The steps for the design are –

Step 1 : Decision for number of flip-flops –

Example : If we are designing mod N counter and n number of flip-flops are required then n can be found out by this equation.

$$N \leq 2^n$$

Here we are designing Mod-10 counter Therefore, N= 10 and number of Flip flops(n) required is

For n =3, $10 \leq 8$, which is false.

For n= 4, $10 \leq 16$, which is true.

Therefore number of FF required is 4 for Mod-10 counter.

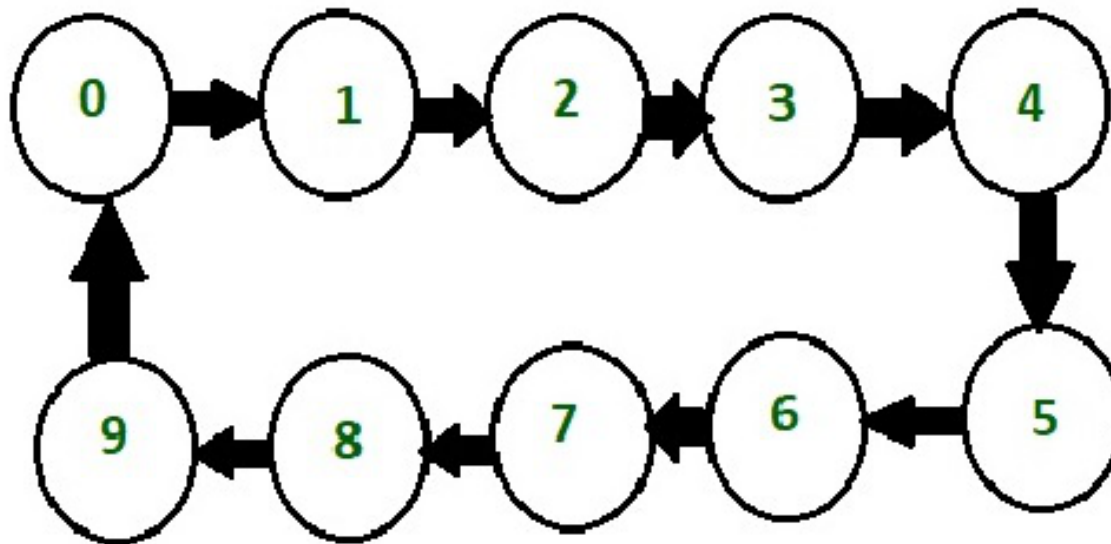
Step 2 : Write excitation table of Flip flops –

Here T FF is used

Previous state(Q_n)	Next state(Q_{n+1})	T
0	0	0
0	1	1
1	0	1
1	1	0

Excitation table of T FF.

Step 3 : Draw state diagram and circuit excitation table –



Counting Sequence of Decade counter

A decade counter is called as mod -10 or divide by 10 counter. It counts from 0 to 9 and again reset to 0. It counts in natural binary sequence. Here 4 T Flip flops are used. It resets after $Q_3 Q_2 Q_1 Q_0 = 1001$.

Circuit excitation table –

Here $Q_3 Q_2 Q_1 Q_0$ are present states of four flip-flops and $Q^*_3 Q^*_2 Q^*_1 Q^*_0$ are next counting state of 4 Flip flops. If there is a transition in current state i.e if Q_3 value changes from 0 to 1 or 1 to 0 then there's corresponding T(toggle) bit is written as 1 otherwise 0.

Q_3	Q_2	Q_1	Q_0	Q^*_3	Q^*_2	Q^*_1	Q^*_0	T_3	T_2	T_1	T_0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	0	0	0	0	1	0	0	1

Circuit excitation table

Step 4 : Create Karnaugh map for each FF input in terms of flip-flop outputs as the input variable –
Simplify the K map –

$Q_3Q_2 \backslash Q_1Q_0$		00	01	11	10
		00	01	11	10
00	0	0	0	0	
01	0	0	1	0	
11	X	X	X	X	
10	0	1	X	X	

$$T_3 = Q_3Q_0 + Q_2Q_1Q_0$$

$Q_3Q_2 \backslash Q_1Q_0$		00	01	11	10
		00	01	11	10
00	0	0	1	0	
01	0	0	1	0	
11	X	X	X	X	
10	0	0	X	X	

$T_2 = Q_1Q_0$

$Q_3Q_2 \backslash Q_1Q_0$		Q_1Q_0			
		00	01	11	10
Q_3	00	0	1	1	0
	01	0	1	1	0
Q_2	11	X	X	X	X
	10	0	0	X	X

$T_1 = Q_3'Q_0$

$M \ Q_3 \backslash$		$Q_2 Q_1$			
		00	01	11	10
M	00	1	1	1	1
	01	1	1	1	1
	11	1	1	1	1
	10	1	1	1	1

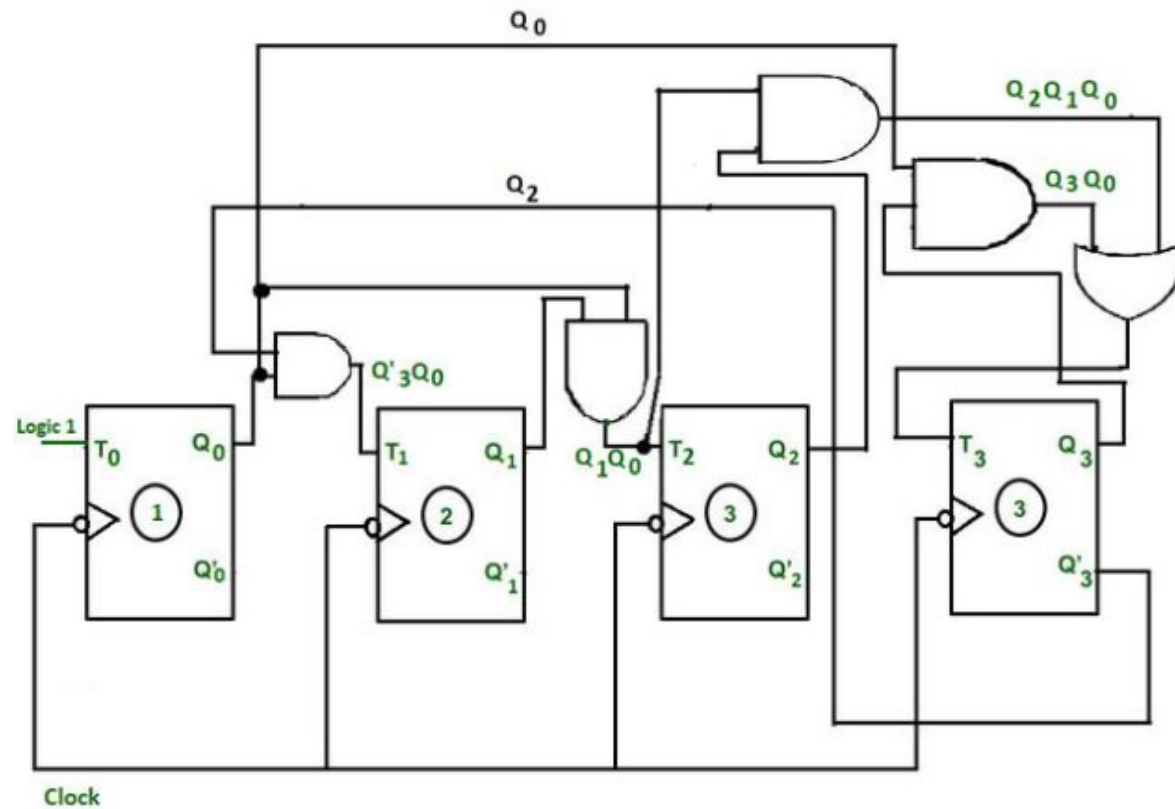
$T_0 = 1$

K map for finding minimal expressions.

Step 5 : Create circuit diagram –

Here negative edge triggered clock is used for toggling purpose.

- The clock is provided to every Flip flop at same instant of time.
- The toggle(T) input is provided to every Flip flop according to the simplified equation of K map.

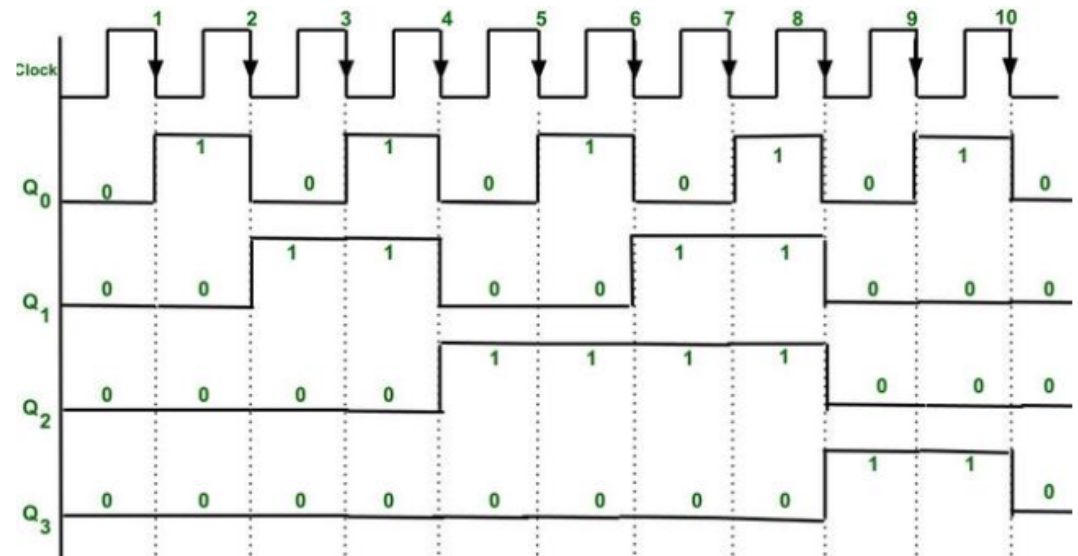


Timing diagram : Here toggling is used.

Previous state(Q_n)	T	Next state(Q_{n+1})
0	0	0
0	1	1
1	0	1
1	1	0

Characteristic table of T FF.

The state of a FF will change only when toggle input(T) of a FF is 1.



Timing diagram of synchronous Decade counter

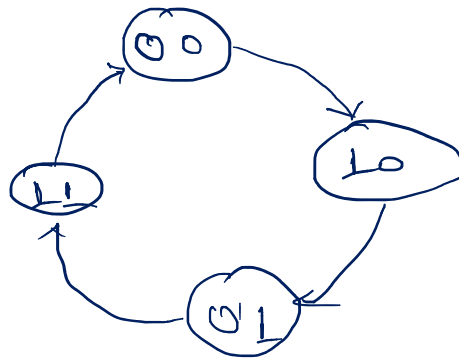
Explanation :

- Initially $Q_3 Q_2 Q_1 Q_0$ are 0 0 0 0.
- The sequence of counter can be verified from the timing diagram. At every falling edge of the clock output Q_0 toggles because T_0 is connected to logic 1.
- T_1 becomes 1 only when expression $T_1 = Q_3' Q_0$ becomes 1 also if clock falling edge occurs (because there is negative edge triggering) then the output state of T_1 i.e Q_1 will change.
- T_2 becomes 1 only when expression $T_2 = Q_1 Q_0$ becomes 1 also if clock falling edge occurs then the output state Q_2 will change.
- T_3 becomes 1 only when expression $T_3 = Q_3 Q_0 + Q_2 Q_1 Q_0$ resultant becomes 1 also if clock falling edge occurs (because there is negative edge triggering) then the state of Q_3 will change.
- We get Output as Q_3 (MSB) $Q_2 Q_1 Q_0$ (LSB).
- After 10th falling edge the output state of all the FFs again becomes 0 0 0 0.

* Synchronous counter Design for given Count sequence.

Q.

$0 \rightarrow 2 \rightarrow 1 \rightarrow 3$



Procedure

- 1) Identify the number of states & FF's
- 2) construct State table.
- 3) Write excitation eqⁿ
- 4) Minimize the logical expression
- 5) Implement the logic circuit.

Soln Step 1:-

No. of states = 4
No. of ff reqd = 2

Step 2:-

State table

PS $Q_1 Q_0$	NS		$D_1 D_0$	
	Q_1^+	Q_0^+	D_1	D_0
0 0	1	0	1	0
1 0	0	1	0	1
0 1	1	1	1	1
1 1	0	0	0	0

Step 3/4

$$D_1 = \overline{Q_1} \overline{Q_0} + \overline{Q_1} Q_0 = \overline{Q_1} (\overline{Q_0} + Q_0) = \overline{Q_1}$$

$$D_0 = \overline{Q_1} Q_0 + Q_1 \overline{Q_0} = Q_1 \oplus Q_0$$

Step 3 :-

CLK

